

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 15

Algorithmic Complexity and the Search for Generative Equations

Assignment

Self-Study Curriculum

Take a dataset from your own work, one where you know (or suspect) the underlying mechanism. Attempt to find the generative equation. You can use any tool: pen and paper, dimensional analysis, symbolic regression software, or manual hypothesis testing.

Write up:

1. The dataset (or a description of it)
2. Your candidate equation(s)
3. How you searched (what your “grammar” was, implicitly or explicitly)
4. How confident you are that you found the *mechanism* vs. a *good fit*
5. What evidence would distinguish a generative equation from a statistical regularity

This is the hardest assignment in the curriculum. If you do not find the equation, describe the search process and explain what made it hard.